CHAPTER 2

INVERSE TRIGONOMETRIC FUNCTIONS

CHAPTER ESSENCE AT A GLANCE:
Principal value branches of inverse trigonometric functions, domains and ranges, and trigonometric identity simplifications.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Principal Value Branches:** Standard trigonometric functions are periodic and many-to-one across the real numbers; restricting their domains to designated principal branches restores strict monotonicity, enabling uniquely defined inverse functions.

• **Symmetry and Negative Arguments:** Inverse sine, inverse tangent, and inverse cosecant are odd functions mapping negative arguments to their corresponding negative angles. Conversely, inverse cosine, inverse secant, and inverse cotangent subtract the acute reference angle from pi.

• **Complementary Angle Pairs:** Across their shared valid domains, the sum of inverse sine and inverse cosine of the same argument equals half pi, as does the sum of inverse tangent and inverse cotangent.

• **Composition within Branch Limits:** The composition of a trigonometric function with its inverse returns the original argument if and only if the angle lies strictly within the principal branch, requiring angle reduction for arguments outside this interval.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$y = \sin^{-1} x \in [-\tfrac{\pi}{2}, \tfrac{\pi}{2}], \quad y = \cos^{-1} x \in [0, \pi]$$

Principal value branches of inverse sine and cosine functions

$$\sin^{-1} x + \cos^{-1} x = \tfrac{\pi}{2}, \quad \tan^{-1} x + \cot^{-1} x = \tfrac{\pi}{2}$$

Complementary angle identities for inverse functions

$$\cos^{-1}(-x) = \pi - \cos^{-1} x, \quad \sin^{-1}(-x) = -\sin^{-1} x$$

Negative argument relations across principal value branches

$$\tan^{-1} x + \tan^{-1} y = \tan^{-1}\left(\tfrac{x+y}{1-xy}\right) \quad (xy < 1)$$

Addition formula for inverse tangent

$$2\tan^{-1} x = \sin^{-1}\left(\tfrac{2x}{1+x^2}\right) = \cos^{-1}\left(\tfrac{1-x^2}{1+x^2}\right)$$

Double angle transformations into inverse sine and cosine

$$3\sin^{-1} x = \sin^{-1}(3x - 4x^3), \quad 3\cos^{-1} x = \cos^{-1}(4x^3 - 3x)$$

Triple angle identities for inverse trigonometric functions

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** [Trap Alert: Domain Exceedance] $\cos^{-1}(\cos(7\pi \text{ over } 6))$ is NOT 7π over 6 because 7π over 6 [0, π]! Rewrite $\cos(7\pi \text{ over } 6) = \cos(2\pi - 5\pi \text{ over } 6) = \cos(5\pi \text{ over } 6)$ implies Result is 5π over 6.

• **Alert:** [Trap Alert: Sine PVB] $\sin^{-1}(\sin(2\pi \text{ over } 3))$ is NOT 2π over 3! Rewrite as $\sin(\pi - \pi \text{ over } 3) = \sin(\pi \text{ over } 3)$ implies Result is π over 3.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. State the domain and principal value branch of all six inverse trigonometric functions. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Principal Value Branches (PVB)]** 1. $y = \sin^{-1} x$: Domain [-1, 1], PVB [-π over 2, π over 2]. 2. $y = \cos^{-1} x$: Domain [-1, 1], PVB [0, π]. 3. $y = \tan^{-1} x$: Domain R, PVB (-π over 2, π over 2). 4. $y = \text{cosec}^{-1} x$: Domain R - (-1, 1), PVB [-π over 2, π over 2] - {0}. 5. $y = \sec^{-1} x$: Domain R - (-1, 1), PVB [0, π] - {π over 2}. 6. $y = \cot^{-1} x$: Domain R, PVB (0, π). [2 Marks]

• **[Boxed Result: Standard Principal Value Branches of all 6 Inverse Trig Functions]**

2. Evaluate: sin[π over 3 - $\sin^{-1}$(-1 over 2)]. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Inner Inverse Evaluation]** $\sin^{-1}$(-1 over 2) = - π over 6. [1 Mark]

• **[Step 2: Outer Trig Evaluation]** sin [π over 3 - $\sin^{-1}$(-1 over 2)] = sin [π over 3 - (- π over 6)] = sin [π over 3 + π over 6] = sin [3π over 6] = sin(π over 2) = 1. [1 Mark]

• **[Boxed Result: sin [π over 3 - $\sin^{-1}$(-1 over 2)] = 1]**

3. Find the principal value of: (a) $\sin^{-1}$(-1 over 2), (b) $\cos^{-1}$(-sqrt(3) over 2), (c) $\tan^{-1}$(-sqrt(3)). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Evaluation of Principal Values]** (a) $\sin^{-1}$(-1 over 2): $\sin^{-1}(-x) = -\sin^{-1}(x)$ for x in [-1, 1]. $\sin^{-1}$(-1 over 2) = - $\sin^{-1}$(1 over 2) = - π over 6 in [-π over 2, π over 2]. [1 Mark]

• (b) $\cos^{-1}$(-sqrt(3) over 2): $\cos^{-1}(-x) = \pi - \cos^{-1}(x)$. $\cos^{-1}$(-sqrt(3) over 2) = π - $\cos^{-1}$(sqrt(3) over 2) = π - π over 6 = 5π over 6 [0, π]. [1 Mark]

• (c) $\tan^{-1}$(-sqrt(3)): $\tan^{-1}(-x) = -\tan^{-1}(x)$. $\tan^{-1}$(-sqrt(3)) = - $\tan^{-1}$(sqrt(3)) = - π over 3 (-π over 2, π over 2). [1 Mark]

• **[Boxed Result: (a) -π over 6 | (b) 5π over 6 | (c) -π over 3]**

4. Find the value of: $\tan^{-1}$(1) + $\cos^{-1}$(-1 over 2) + $\sin^{-1}$(-1 over 2). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Evaluate Each Term]** 1. $\tan^{-1}$(1) = π over 4 (since tan(π over 4) = 1 and π over 4 in (-π over 2, π over 2)). 2. $\cos^{-1}$(-1 over 2) = π - $\cos^{-1}$(1 over 2) = π - π over 3 = 2π over 3. 3. $\sin^{-1}$(-1 over 2) = - $\sin^{-1}$(1 over 2) = - π over 6. [2 Marks]

• **[Step 2: Sum]** Value = π over 4 + 2π over 3 - π over 6 = (3π + 8π - 2π) over 12 = 9π over 12 = 3π over 4. (Alternative: $\cos^{-1}$(-1 over 2) + $\sin^{-1}$(-1 over 2) = π over 2 by identity; value = π over 4 + π over 2 = 3π over 4). [1 Mark]

• **[Boxed Result: $\tan^{-1}$(1) + $\cos^{-1}$(-1 over 2) + $\sin^{-1}$(-1 over 2) = 3π over 4]**

5. Evaluate: $\cos^{-1}$(cos(7π over 6)) and $\sin^{-1}$(sin(2π over 3)). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Evaluation within Principal Value Branches]** (a) $\cos^{-1}$(cos(7π over 6)): 7π over 6 not in [0, π]. Rewrite: cos(7π over 6) = cos(2π - 5π over 6) = cos(5π over 6). Since 5π over 6 in [0, π]: $\cos^{-1}$(cos(7π over 6)) = $\cos^{-1}$(cos(5π over 6)) = 5π over 6. [1.5 Marks]

• (b) $\sin^{-1}$(sin(2π over 3)): 2π over 3 [-π over 2, π over 2]. Rewrite: sin(2π over 3) = sin(π - π over 3) = sin(π over 3). Since π over 3 [-π over 2, π over 2]: $\sin^{-1}$(sin(2π over 3)) = $\sin^{-1}$(sin(π over 3)) = π over 3. [1.5 Marks]

• **[Boxed Result: $\cos^{-1}$(cos(7π over 6)) = 5π over 6 and $\sin^{-1}$(sin(2π over 3)) = π over 3]**

6. Find the value of x if: $\tan^{-1}\left(\frac{(x-1)}{(x-2)}\right) + \tan^{-1}\left(\frac{(x+1)}{(x+2)}\right) = \frac{\pi}{4}$. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Apply $\tan^{-1}$ A + $\tan^{-1}$ B Formula]** $\tan^{-1}$ A + $\tan^{-1}$ B = $\tan^{-1}$ [(A + B) over (1 - A B)]. Here A = (x - 1) over (x - 2) and B = (x + 1) over (x + 2). A + B = [(x-1)(x+2) + (x+1)(x-2)] over (x^2 - 4) = (x^2 + x - 2 + x^2 - x - 2) over (x^2 - 4) = ($2x^2$ - 4) over (x^2 - 4). 1 - A B = 1 - [(x-1)(x+1) over (x-2)(x+2)] = 1 - (x^2 - 1) over (x^2 - 4) = (x^2 - 4 - x^2 + 1) over (x^2 - 4) = -3 over (x^2 - 4). [2 Marks]

• **[Step 2: Equating and Solving]** (A + B) over (1 - AB) = ($2x^2$ - 4) over (-3). $\tan^{-1}$ [($2x^2$ - 4) over (-3)] = π over 4 ⇒ ($2x^2$ - 4) over (-3) = tan(π over 4) = 1. $2x^2$ - 4 = -3 ⇒ $2x^2$ = 1 ⇒ x^2 = 1 over 2 ⇒ x = ± 1 over √2. [2 Marks]

• **[Boxed Result: x = ± 1 over √2]**

7. Write in simplest form: $\tan^{-1}\left(\frac{(cosx - sinx)}{(cosx + sinx)}\right)$, 0 < x < π. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Divide Numerator and Denominator by cos x]** (cos x - sin x) over (cos x + sin x) = (1 - tan x) over (1 + tan x). [1.5 Marks]

• **[Step 2: Use tan(A - B) Identity]** Substitute 1 = tan(π over 4): (tan(π over 4) - tan x) over (1 + tan(π over 4) tan x) = tan(π over 4 - x). $\tan^{-1}$ [tan(π over 4 - x)] = π over 4 - x. [1.5 Marks]

• **[Boxed Result: Simplest Form = π over 4 - x]**